DESIGN 05 OF 10

NEURO-TREAD

A raised neural network over a dimpled adaptive base.

THE DESIGN

A network of 48 neurons joined by curved axon ridges rises to the full 3.5 mm crest, standing proud of a field of small bumps. The ridges guide the fingers while the bump field adds fine grip.



CLOSE-UP · MID BODY

57.8 g
PLA, SEALED CORE

48
NODES

3.50 mm
NETWORK HEIGHT



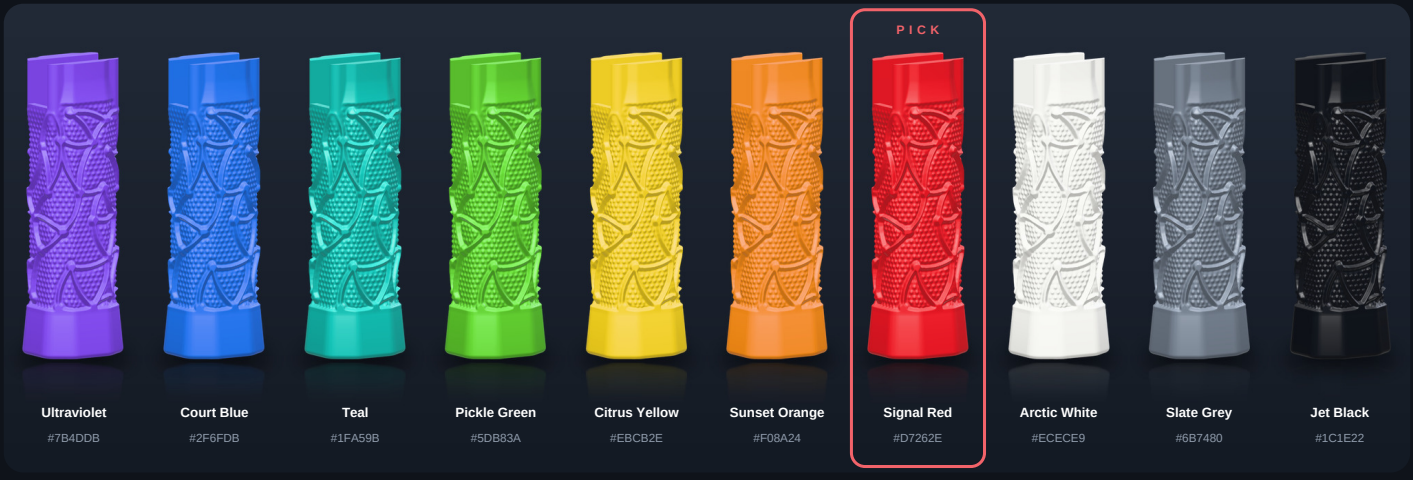
THE HANDLE

A sleeve that slides over your paddle's flat handle: a 30.2 × 16.2 mm slot (the measured 30 × 16 mm tongue plus 0.2 mm), 130 mm deep so the paddle seats fully, with a thin 3.4 mm base and a counterbored 3.5 mm hole that takes a low-head M3 cap screw flush. The head is solid, with both short side walls of the slot slit open at the top so the flared throat of the paddle slides in smoothly; a sealed hollow core lightens the rest. The outer shape is the same on all ten designs.

133.4 mm	30 × 16 mm	130 mm	M3 low-head
LENGTH	PADDLE SLOT	SLOT DEPTH	SCREW, FLUSH

CHOOSE YOUR COLOUR

DESIGNER'S PICK: SIGNAL RED



Neuro-Tread, all round.

Four views, four surfaces and the idea behind the design. Shown as it prints: one solid PLA colour.



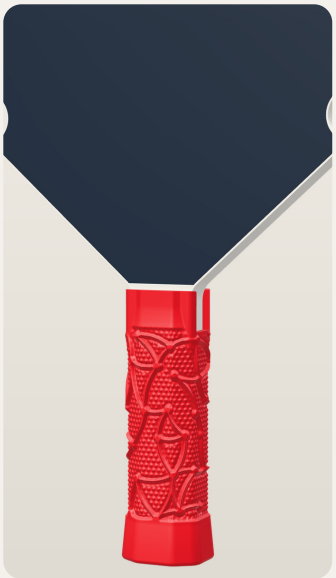
3/4 VIEW · TURN 0°



SIDE VIEW · TURN 90°



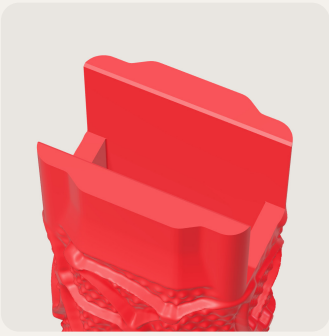
CUTAWAY · HOLLOW CORE



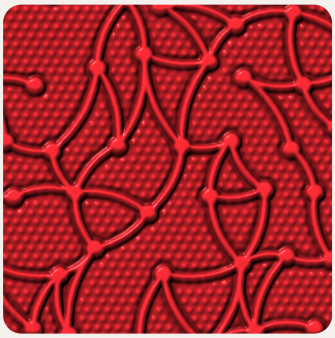
ON A PADDLE · ILLUSTRATION



CLOSE-UP · MID BODY



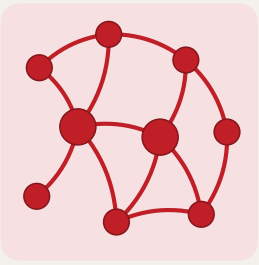
RIM & SLOT · FROM ABOVE



UNROLLED · LIT RELIEF



BUTT · Ø3.5 HOLE + CBORE



DESIGN INSPIRATION

Neural networks and leaf veins: branching lines that meet at nodes. The ridges read like a map under the palm, while the fine bumps between them work like grip tape.

WHY IT GRIPS

- 48 nodes (8 × 6) joined by curved 1.5 mm ridges
- Bump field on a 2.66 mm hex pitch, 1.4–2.4 mm high
- Network and nodes reach the full 3.5 mm crest

PRINT TIP

Ridges and bumps are gentle slopes, printed in one material.

THE WHOLE COLLECTION



The paddle in the illustration is generic, drawn with your 30 × 16 mm handle. Every design prints in any of the ten PLA colours shown on page 1. 100 % infill · 0.20 mm layers · upright, butt end down.